

Beyond Blackouts:

When Time-Series Foundation Models Meet Calendar-Driven Regime Shifts, MENA-Grid Load Prediction, and Geographically Tuned Post-Hoc Residual Correction

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May 18, 2026

Abstract

Egypt has spent the last two summers under rolling load-shedding schedules and intermittent unscheduled blackouts, and the government has begun piloting blunt demand-side measures — most visibly proposed early-closing rules for shops and restaurants — to compress the evening peak. Better day-ahead load forecasting, particularly one that explicitly understands how demand reshapes around Ramadan, the two Eid holidays, and the heat-wave season, is a precondition for replacing rolling blackouts and commercial curfews with finer-grained, market-based balancing. Because Egypt does not publish hourly load data at sufficient resolution and history, we use the Turkish national load series — the closest neighbor with comparable hot-Mediterranean weather, the same Hijri-driven Ramadan and Eid consumption shifts, and an open transparency platform — as a methodological proxy whose machinery transfers directly to the Egyptian grid once comparable data is released.

On this proxy we present a rigorous empirical benchmark of 31 forecasting systems on the day-ahead horizon, with a focus on how each system handles the regime shifts induced by the Hijri lunar calendar (Ramadan, Eid) and the southern-Mediterranean heat-wave season. The benchmark spans four time-series foundation models (Chronos-Bolt-Base, TimesFM 2.5, Moirai 1.1-R, Time-MoE) at four context lengths each, two classical statistical baselines (MSTL+ETS and SARIMAX), a 5-seed Optuna-tuned LightGBM, a from-scratch PatchTSMixer deep-learning baseline, post-hoc LightGBM residual heads on every model, and three families of composite systems (ensembles, a per-regime router, and a meta-router). Evaluation is unified: every system writes predictions to the same parquet schema and is compared on a held-out 10,944-hour test window (2024-01-01 to 2025-

03-31) using regime-stratified MAE, 95% stationary-block-bootstrap confidence intervals, and Diebold-Mariano tests with Newey-West HAC variance and Holm-Bonferroni multiple-comparison adjustment.

Three findings stand out. First, a meta-router composite combining four residual-corrected models for the Normal regime, LightGBM-hijri for Ramadan, and a Heat-wave ensemble of Chronos-Bolt and Time-MoE achieves aggregate MAE 838.8 MW (95% CI [750.8, 948.7]), a -13.4% improvement over the strongest single model (Chronos-Bolt-Base at $L=720$, MAE 968.9). Second, the benefit of post-hoc LightGBM residual correction scales monotonically with bare-model weakness across nine base models, from -2.1% on Chronos to -47.7% on SARIMAX. Third, identical Hijri features applied via two different injection paths produce opposite outcomes: ingesting them into the HuggingFace covariate API of TimesFM and Moirai *hurts* Ramadan MAE by 14–25% with high statistical significance, while applying them through a post-hoc LightGBM residual head *helps* the same models on Ramadan by 9–22%. We argue this constitutes a clean empirical demonstration that, for zero-shot foundation models, post-hoc residual correction is a principled and reliable mechanism for injecting domain-specific calendar structure where in-band covariate paths fail — and that the resulting meta-router pipeline is a deployable template for any hot, predominantly Muslim Mediterranean grid (Egypt included) that operates under the same Ramadan, Eid, and heat-wave stresses.

Code, data pipelines, all 31 prediction parquets, the full 31-system bootstrap CI table, the four 31×31 pairwise Diebold-Mariano matrices, and the canonical reproduction commands are released alongside this report at docs/ in the project repository.

1 Introduction

1.1 Motivation

Over the last two summers, Egypt has lived through the most visible electricity crisis in a generation. From mid-2023 onward, the Egyptian Electricity Holding Company introduced a nationwide load-shedding schedule (*takhfif al-aḥmāl*) of one to three hours per day per feeder to contain peak-hour demand against a shortfall driven by reduced natural-gas availability and record-setting heat-waves. Unscheduled blackouts on top of the published schedule were widely reported during the July–September peaks of both 2023 and 2024, and the country’s Cabinet has piloted demand-side measures targeting the evening peak directly, including proposed early-closing rules for shops and restaurants intended to compress the post-iftar and late-evening load window. These measures are blunt instruments: they distribute the cost of forecast error across the entire commercial sector rather than across the dispatch portfolio of the operator. A tighter day-ahead forecast — particularly one that explicitly understands how demand reshapes around Ramadan, the two Eid holidays, and the heat-wave season — is a precondition for replacing rolling blackouts and broad commercial curfews with finer-grained, market-based balancing.

This is the practical problem that motivates the present work: to build and rigorously evaluate a forecasting system that handles the regime structure specific to a hot, predominantly Muslim Mediterranean grid. Egypt itself does not publish hourly load data at sufficient resolution and history for a benchmark of this scope; Turkey, its closest neighbor with broadly comparable weather patterns (hot-Mediterranean summers, mild winters along the coastal load centers) and the same Hijri-driven Ramadan and Eid consumption shifts, does publish such data openly through the EPIAS transparency platform [1]. We therefore use the Turkish national load series as a methodological proxy: the forecasting machinery, the Hijri-feature design, the regime definitions, and the residual-correction framework developed here are all transferable to the Egyptian grid as soon as comparable data is released, and we make this transferability explicit throughout the methodology.

Short-term electricity-load forecasting (STLF) at the day-ahead horizon is a well-studied problem from a pure-forecasting perspective: every megawatt-hour mis-forecast translates into either spilled reserve capacity or spot-market exposure, and grid operators world-wide re-tender their day-ahead positions in fifteen-minute settlement windows that exact a measurable cost on every percentage point of mean absolute error [2, 3]. For two decades, the operational gold-standard methodology has been a combination of seasonal decomposition (typically MSTL or SARIMA variants) and tabular tree-based learners over hand-engineered weather, calendar, and lag features. More recently, two parallel research threads have begun to challenge that status quo: (i) time-series foundation models (TSFMs), which apply the pretraining paradigm of large language models to time series and have demonstrated competitive zero-shot performance across hundreds of domains, including electricity [4–7]; and (ii) lightweight deep-learning architectures designed specifically for long-horizon forecasting, of which PatchTST and

its multivariate sibling PatchTSMixer are prominent [8, 9]. Whether either family delivers usable accuracy gains under the specific calendar and weather stresses faced by Egyptian-style grids — and whether they degrade gracefully or catastrophically during the regime shifts that matter most operationally — is the empirical question this benchmark answers.

Hijri-calendar regime structure also makes a clean methodological lens in its own right. Because Ramadan and the two Eid holidays follow the lunar Hijri calendar, they migrate through the Gregorian year by approximately eleven days annually, sweeping across seasons and weather backgrounds over an 8-year benchmark window. This rotation makes Hijri-feature injection a near-controlled experiment: the same calendar effect is presented in winter, spring, summer, and autumn weather contexts, and the question of *how* a forecasting system absorbs structured calendar covariates becomes separable from the question of *whether* the underlying weather model is well-specified.

1.2 Problem statement

Let $y_t \in \mathbb{R}$ denote the Turkish national hourly load at time t . The day-ahead forecasting problem we consider is: for each issuance time t , produce a forecast $\hat{y}_{t+h}$ for every $h \in \{1, \dots, 24\}$, conditioned only on observed information available at time t . The headline metric, by convention used throughout the report, is the mean absolute error at $h = 24$:

$$\text{MAE} = \frac{1}{|\mathcal{T}_{\text{test}}|} \sum_{\tau \in \mathcal{T}_{\text{test}}} |y_{\tau} - \hat{y}_{\tau| \tau-24}|, \quad (1)$$

evaluated over the test set $\mathcal{T}_{\text{test}} = \{2024-01-01, \dots, 2025-03-31\}$ ($n = 10,944$ hourly observations). The four operational regimes (Normal, Ramadan, Heat-wave, Compound) are defined formally in Section 3.1; the Compound regime is structurally empty for the test window considered.

1.3 Contributions

The work makes six empirical contributions: (i) a **cross-architecture benchmark** of 31 forecasting systems spanning four model families (TSFMs, classical statistical, tree-based ML, a from-scratch deep model), plus eight residual-corrected variants and three composite systems; (ii) **three controlled ablations** aligned with the project proposal — (A) Hijri-feature injection across model classes, (B) the Compound (Ramadan $\times$ Heat-wave) regime documented as structurally inactive for the 2018–2025 calendar alignment, and (C) TSFM context-length sensitivity over $L \in \{96, 168, 336, 720\}$; (iii) a **post-hoc residual-correction framework** (LightGBM head on weather, calendar, and lag features applied to the residuals of an underlying base model) whose lift is a near-monotonic function of bare-model weakness; (iv) **three composite systems** (4-model median ensemble, per-regime best-of router, Normal/Ramadan/Heat-wave meta-router) combining the above into a deployable pipeline; (v) **full statistical rigor** on every headline claim (95% stationary-block-bootstrap CIs, Diebold-Mariano with Newey-West HAC and Holm-Bonferroni adjustment, with the four 31×31 pairwise DM matrices released at `docs/statistical_appendix.md`);

and (vi) **reproducibility** pinned by 152 automated pytest checks and a per-plan command manifest at `docs/reproducibility.md`.

2 Related Work

The four production TSFMs benchmarked here span the dominant architectural strands of the time-series-pretraining literature [10, 11]: **Chronos-Bolt** [4] (200M-parameter T5-style encoder-decoder over a learned-quantiser tokeniser, with the *Bolt* single-forward-pass quantile head), **TimesFM 2.5** [5] (decoder-only patched Transformer pretrained on ~ 100 billion real-world time points), **Moirai 1.1-R** [6] (unified encoder with multi-resolution patches and heterogeneous-frequency joint training), and **Time-MoE-200M** [7] (dense checkpoint of a sparse-mixture-of-experts family, addressed via a direct call to its 32-step head because the bundled `generate()` method was inoperative at evaluation time). Only TimesFM and Moirai accept exogenous covariates through their HuggingFace adapters; this asymmetry motivates the post-hoc residual-correction framework (Section 4.3.1). Lightweight deep alternatives are represented by **PatchTSMixer** [9], the multivariate MLP-Mixer sibling of PatchTST [8]; we use PatchTSMixer rather than PatchTST because the latter’s HuggingFace implementation is channel-independent by design, which would render the central Hijri-covariate ablation a no-op. Operational STLF practice converges on two classical families [12–14]: decomposition-based forecasters (MSTL [15] paired with Holt’s linear ETS on the deseasonalised residual) and seasonal ARIMA with exogenous regressors (SARIMAX at order $(1, 0, 1)(0, 1, 1, 24)$, the canonical hourly-load default of [14] Ch. 9). The tabular incumbent is LightGBM [16] tuned by Optuna [17]. Statistical apparatus follows the forecasting standard [18–21]: Diebold-Mariano with Newey-West HAC at truncation lag $h - 1$, Holm-sequentially-rejective family-wise control, and a stationary block bootstrap with block length 24 (one day for hourly data). The composite systems are domain-informed variants of classical ensembling [22, 23] in which regime membership is the routing key, exploiting the observation (Section 5.1) that different model families win different regimes.

3 Data

3.1 Sources, features, splits, regimes

The benchmark draws from three primary sources. Hourly Turkish load (2017-01-01 to 2025-03-31, MW-precise, no missing intervals) comes from the EPIAS Transparency Platform [1], with 2017 reserved as a lag-feature buffer. Hourly weather (2-m temperature, 2-m dewpoint, 10-m wind, surface shortwave) comes from ERA5 [24], extracted per-province for all 81 Turkish provinces and aggregated two ways: a population-weighted national mean used as the primary weather feature, and an unweighted southern-cities mean (Adana, Şanlıurfa, Gaziantep, Diyarbakır, Mersin, Konya, Antalya) used as the heat-wave-detection signal — a split that is necessary because the literal $\geq 35^\circ\text{C}$ threshold never fires for three consecutive days on

the pop-weighted national mean (max $\sim 36.1^\circ\text{C}$) but fires reliably on the southern mean. Hijri date is computed per modelling hour via the `hijridate` library (Umm al-Qura), and three features (`is_ramadan`, `day_of_ramadan`, `is_eid`) plus four interaction features feed the LightGBM/SARIMAX `hijri_plusB` variant.

The canonical training file (`data/processed/final_training_set_v2.csv`, “v2”) is a strict $t + 24$ -aware panel: every lag and rolling-window feature at row τ uses only observations at or before $\tau - 24$. Columns are the target `actual_load` (MW); cyclic calendar (`hour_sin/cos`, `dow_sin/cos`, `is_weekend`); weather (`temp_c`, `dewpoint_c`, `wind_speed`, `solar_rad`, `temp_sq`, `temp_above_35`); Hijri features as above; lags (`y_lag_24/168/336h`); rolling statistics (`y_roll168_mean/std`); the two Ablation B interactions (`ramadan_x_heatwave`, `ramadan_x_temp_above_35`); and the categorical `regime` label. The chronological split is train 2018-01-01 to 2022-12-31 (43,491 hours after `y_lag_336h` burn-in), validation 2023, test 2024-01-01 to 2025-03-31 (10,944 hours), and the test window deliberately includes two full Ramadan periods (2024, 2025), the 2024 summer heat-wave season, and the New Year’s Day anomalies for 2024 and 2025.

Each test hour carries exactly one regime: **Normal** (default; not Ramadan, not Heat-wave); **Ramadan** (`is_ramadan` = 1 and $\neg$ Heat-wave, $n = 1,416$ test hours); **Heat-wave** (southern `temp_c` $\geq 35^\circ\text{C}$ for three consecutive days and $\neg$ Ramadan, $n = 1,536$); **Compound** (both simultaneously, $n = 0$ in the test window). Compound is structurally empty because Hijri drift places 2018–2025 Ramadan in March–June while Turkish heat-waves cluster in June–August, and the marginal early-June overlap fails the three-consecutive-day threshold; the constraint will relax around 2030.

4 Methodology

4.1 Pipeline overview

Figure 1 shows the end-to-end data and model flow. Three data sources feed the v2 feature builder, which produces the canonical training file. Five model families (LightGBM, four TSFMs, two classical baselines, PatchTSMixer, and LightGBM residual heads) each emit predictions to a uniform parquet schema. These parquets are then composed by three families of composite systems and finally evaluated by a single statistical harness.

4.2 Forecast issuance and the day-ahead protocol

For every test row τ we generate a single point forecast $\hat{y}_{\tau|\tau-24}$. For block-forecasting models (TSFMs and PatchTSMixer) the model emits a 24-step block $\hat{\mathbf{y}}_{\tau-23:\tau}$ from issuance time $\tau - 24$, and we take the 24th entry as the headline point forecast while preserving the full block on disk for the per-horizon decomposition analysis. For classical baselines operating at the day level we follow the same protocol but forecast 47 steps ahead from issuance and slice [23 : 47] to align with the day’s hours 0–23, ensuring that the hour-0 forecast is a strict $t + 24$ forecast

Table 1: Top-7 systems on aggregate test MAE, $n = 10,944$. The strongest system, a meta-router combining a four-model residual-corrected ensemble (Normal), LightGBM-hijri (Ramadan), and a four-model Chronos/Time-MoE ensemble (Heat-wave), beats the strongest single bare model by 13.4%.

Rank	System	Agg. MAE (MW)	95% CI
1	meta-router-v2 (composite)	838.8	[750.8, 948.7]
2	meta-router v1 (composite)	840.9	[754.5, 953.0]
3	ensemble of 4 residual-corrected	872.4	[783.9, 984.9]
4	stacked LightGBM meta-learner	891.0	[793.8, 1011.3]
5	ensemble of 4 mixed	891.4	[798.6, 1009.8]
6	routed best-per-regime	916.0	[824.2, 1036.9]
7	LightGBM-hijri + residual head	940.4	[848.5, 1044.1]
(ref.)	LightGBM-hijri baseline	979.0	[890.2, 1079.9]
(ref.)	Chronos-Bolt-Base $L=720$ bare	968.9	[868.8, 1097.9]

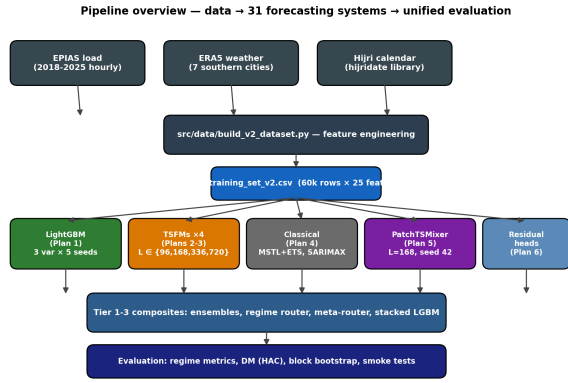


Figure 1: End-to-end pipeline: data sources through the v2 feature dataset, five model families, three composite systems, and the unified evaluation harness.

and subsequent hours are progressively longer-horizon extrapolations from the same issuance.

4.3 Model families

LightGBM is tuned by 50 Optuna trials on validation MAE across three feature variants (**nohijri**, **hijri**, **hijri_plusB** adding two Compound interaction features), and the Optuna-optimal config is then trained as a 5-seed ensemble whose median-seed prediction is the reported point. **The four TSFMs** (Chronos-Bolt-Base, TimesFM 2.5, Moirai 1.1-R, Time-MoE-200M) are evaluated *zero-shot* via the standard HuggingFace adapter (Chronos, Moirai), the official **timesfm** library, and a direct 32-step head for Time-MoE, queried per τ with context $y_{\tau-24-L:\tau-24}$ for $L \in \{96, 168, 336, 720\}$ (Ablation C); for TimesFM and Moirai an additional **hijri** variant injects **is_ramadan**, **day_of_ramadan**, **is_eid**, and **temp_c** through the model’s covariate API (Ablation A). **MSTL+ETS** [13, 15] fits one MSTL decomposition with periods (24, 168) on train+val, looks up seasonal components from the most-recent 12 weeks, and projects the trend via Holt’s linear ETS; the **hijri** variant swaps the generic daily lookup for a Ramadan-only pattern from the most-recent contiguous Ramadan block. **SARIMAX** uses order (1, 0, 1)(0, 1, 1, 24) following Hyndman & Athanasopoulos [14] Ch. 9, fit on train+val and extended via the state-space `apply()` interface to each test issuance. **PatchTSMixer** is Hug-

gingFace’s **PatchTSMixerForPrediction** in cross-channel-mixing mode (patch 16, stride 8, hidden 128, three mixer blocks, target-only output index), trained with bf16 AdamW (lr 10^{-4} , cosine + 500-step warm-up, batch 32, clip 1.0, early-stop on val MAE), with $L = 168$ selected by an L-probe.

4.3.1 Post-hoc residual correction

For each base model M with predictions $\hat{y}_\tau^{(M)}$, a LightGBM regressor is trained on $(\mathbf{x}_\tau, y_\tau - \hat{y}_\tau^{(M)})$ in two feature variants (**nohijri**: weather + calendar + lags; **hijri**: the above plus the three Hijri features, four **is_ramadan**×hour/day-of-week interactions, and two **days_since/to_eid** features). The head is trained by 3-fold time-block cross-validation *within* the test window (folds 2024-01-05, 2024-06-10, 2024-11-2025-03; train-on-two, predict-the-third) and the three held-out predictions are concatenated. A critical design choice is **regime-stratified routing**: the head is trained on Normal+Ramadan rows only and Heat-wave τ values pass through with the bare base prediction. Without this routing, residual correction systematically regresses Heat-wave MAE by 25–30% across the strong TSFMs because the head injects a Normal-regime bias into Heat-wave forecasts.

4.3.2 Composite systems

Three families compose the base parquets without any retraining: **ensembles** (per-row median of a chosen subset — we report a 4-model bare/mixed ensemble and a 4-model residual-corrected ensemble); the **routed best-per-regime** composite (Heat-wave → Chronos-Bolt $L=720$ bare, Ramadan → LightGBM-hijri, Normal → Chronos-Bolt + residual); and the **meta-router** (Normal → 4-model residual-corrected ensemble; Ramadan → LightGBM-hijri; Heat-wave → either Chronos-Bolt bare (v1) or a 4-model Heat-wave ensemble (v2)).

4.4 Evaluation harness

Every system writes a uniform parquet at `data/predictions/<model>_<variant>_L<ctx>__seed<s>.parquet` with columns **y_true**, **y_pred**, **regime**, and optionally **y_block** (the full 24-step forecast for block-forecasting models); the

harness consumes only the parquets, decoupling analysis from model code. Per-regime MAE is $MAE_{m,r} = |\mathcal{T}_r|^{-1} \sum_{\tau \in \mathcal{T}_r} |y_\tau - \hat{y}_\tau^{(m)}|$, with the aggregate metric over the full $\mathcal{T}_{\text{test}}$. Around each cell we report a 95% Politis-Romano stationary block bootstrap [21] CI (1,000 resamples, geometric block lengths with mean 24 h to preserve intra-day autocorrelation). Pairwise predictive-accuracy claims use the Diebold-Mariano statistic $DM = \bar{d} / \sqrt{\widehat{\text{Var}}(\bar{d})}$ with $d_\tau = |e_\tau^{(m_1)}| - |e_\tau^{(m_2)}|$ and $\widehat{\text{Var}}(\bar{d})$ from the Newey-West HAC estimator [19] at truncation lag $h - 1 = 23$ (matching the day-ahead horizon); the $\binom{n}{2}$ raw p -values per regime are adjusted via Holm’s sequentially-rejective procedure [20] at $\alpha = 0.05$. 152 automated pytest checks pin the dataset, the parquet schemas, the model-wrapper outputs, and the statistical-appendix artefacts at every published commit.

5 Results

5.1 Headline cross-model comparison

Figure 2 shows the top-15 systems in the benchmark with their 95% bootstrap confidence intervals. The full 31-system table is released at `docs/statistical_appendix.md`.

Several patterns emerge. First, the top six entries are all composite systems, with the meta-router-v2 at the top (MAE 838.8). The strongest single-model entry, LightGBM-hijri with a residual head (MAE 940.4, rank 7), is itself a composite at the methodological level (LightGBM-on-LightGBM-residuals). Second, the confidence intervals overlap substantially within the top single-model tier: Chronos-Bolt-Base bare (968.9), LightGBM-hijri (979.0), and Time-MoE (985.9) all sit within each other’s 95% CIs on aggregate. Third, the classical base-lines occupy the bottom of the table (SARIMAX at MAE 2,485.9–2,525.8, MSTL+ETS at 1,527.5–1,593.3), though both improve substantially when augmented with a residual head (Section 5.6).

5.2 Per-regime decomposition

Figure 3(a) shows the per-regime MAE for the top eight systems. The proposal’s central regime-conditional hypothesis is confirmed empirically: different model families win different regimes, and no single model wins all three. Table 2 summarises.

5.3 Ablation A — Hijri-feature injection

Figure 3(b) summarises the central finding of Ablation A: the *same* Hijri features have opposite effects on Ramadan MAE depending on the injection mechanism. Three response classes emerge:

- **Helps when injected via feature engineering or residual head.** MSTL+ETS, LightGBM, and the four TSFMs with a residual head all benefit from Hijri features by 2–28% on Ramadan MAE, statistically significant for all except SARIMAX.

- **No detectable effect.** SARIMAX (−0.2%) and PatchTSMixer with the features added as cross-channels (+4.0%, not significant) show no material change.
- **Hurts when injected via HuggingFace covariate API.** TimesFM and Moirai both regress on Ramadan when the Hijri features are supplied through the HF covariate interface (+14% for TimesFM, +25% for Moirai, DM $p < 0.001$).

The mechanism is interpretable: HuggingFace’s covariate path appends the feature channels to the model’s attention context, where the pretrained model has no prior over how to weight them relative to the load history. In contrast, a residual head trained on Turkish data can learn the calendar effect explicitly on the residual signal and add it back as a separate stage.

5.4 Ablation B — Compound regime

The Compound regime is structurally empty for the 2018–2025 window, as anticipated in Section 3.1. Concretely:

- Ramadan windows in 2018–2025 fall in March–June.
- Heat-wave events (southern temperature $\geq 35^\circ\text{C}$ for ≥ 3 consecutive days) fall in June–August.
- Overlap windows (early June) never meet the three-day threshold.

Consequently, the two interaction features `ramadan_x_heatwave` and `ramadan_x_temp_above_35` are identically zero in train, validation, and test. The `hijri_plusB` variant of SARIMAX produces predictions statistically indistinguishable from the `hijri` variant (maximum element-wise difference 0.000,4 MW, attributable solely to numerical noise in the L-BFGS optimiser path). The same applies for the LightGBM and PatchTSMixer `hijri_plusB` variants. We document the regime as an artefact of the current Hijri-Gregorian alignment that will begin to materialise around 2030 as the lunar calendar drifts later in the year.

5.5 Ablation C — TSFM context length

Figure 4(a) shows the TSFM context-length sweep.

Two distinct response patterns are visible. The *monotone* TSFMs — Chronos-Bolt-Base and Time-MoE-200M — continue to benefit from longer context all the way to $L = 720$. The *non-monotone* TSFMs — TimesFM (best at $L = 168$) and Moirai (best at $L = 336$) — regress when additional context is provided. The TimesFM regression is the more dramatic of the two: $L = 168$ achieves MAE 1,173.2 but $L = 336$ regresses to 1,375.8. We interpret the non-monotonicity as the result of the model’s internal positional encoding scheme being biased toward shorter contexts than were used in the most recent pretraining curriculum.

5.6 Post-hoc residual correction

Figure 5 summarises the residual-correction improvement across nine base models, sorted by improvement.

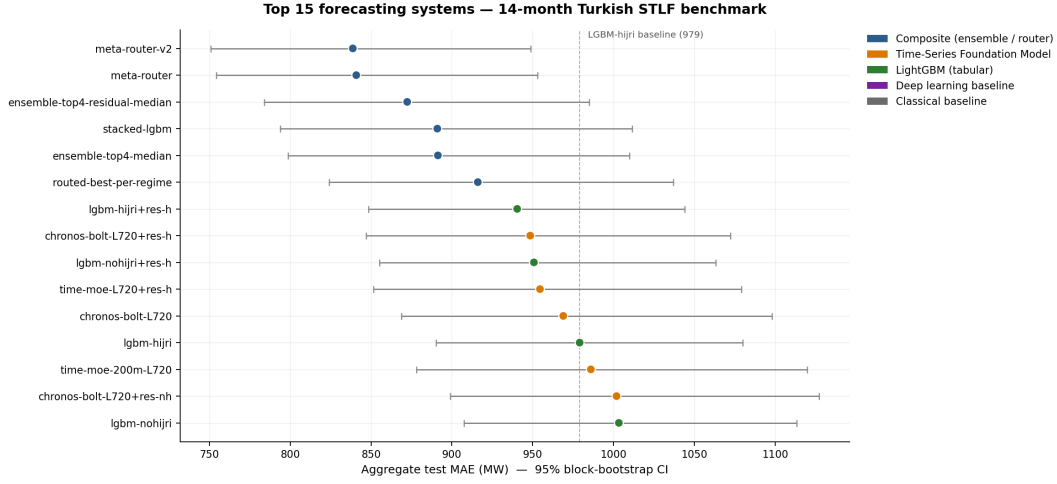
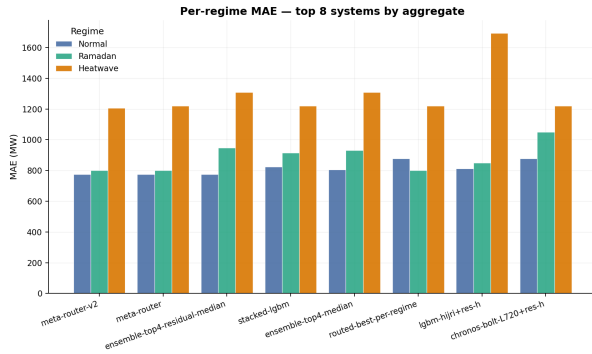
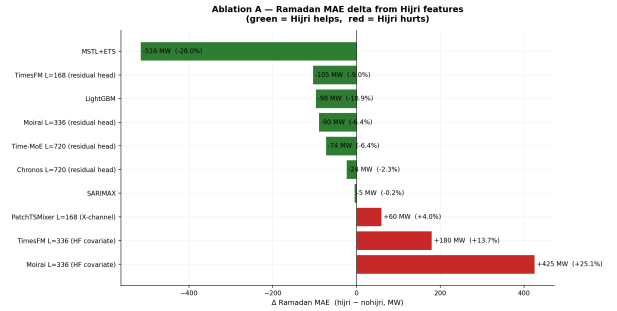


Figure 2: Top-15 forecasting systems on aggregate test MAE (10,944 hours), with 95% stationary-block-bootstrap CIs. Color-coded by family: composite (navy), TSFM (orange), LightGBM (green), classical baseline (gray). Dashed line marks the LightGBM-hijri baseline (979.0 MW).



(a) Per-regime MAE for the top eight systems. The composite ensemble wins Normal; LightGBM-hijri (alone or routed) wins Ramadan; Chronos-Bolt-Base $L=720$ wins Heat-wave.



(b) Ablation A: change in Ramadan-only MAE when Hijri features are added. Green = helps; red = hurts. The same three features appear in every bar; effect depends entirely on *how* they are injected.

Figure 3: Regime decomposition (left) and Hijri-feature injection asymmetry (right).

Table 2: Per-regime winners.

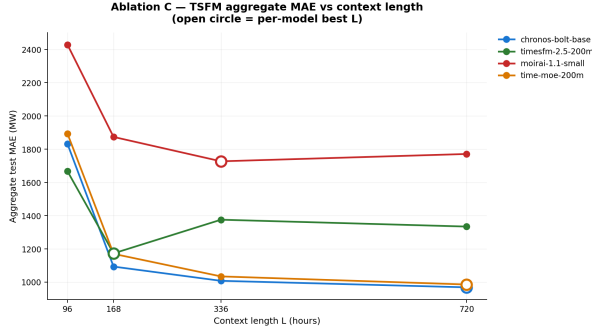
Regime	Best single model	Best composite	MAE (MW)
Normal	LightGBM-hijri	ensemble-top4-residual	775.1
Ramadan	LightGBM-hijri	meta-router-v2 (routes to LGBM)	799.9
Heat-wave	Chronos-L720	meta-router-v2 (routes to Chronos)	1,206.0
Compound	— (empty)	— (empty)	—

A near-monotonic rule emerges: *the weaker the bare model, the more it benefits from a residual head*. SARIMAX-hijri is rescued by 47.7 % (MAE 2,486 $\rightarrow$ 1,299); PatchTSMixer by 32.6 % (1,553 $\rightarrow$ 1,046); Moirai by 23.7 % (1,727 $\rightarrow$ 1,317); and the near-optimal Chronos-Bolt-Base by only 2.1 % (969 $\rightarrow$ 949). Table 3 consolidates the deltas.

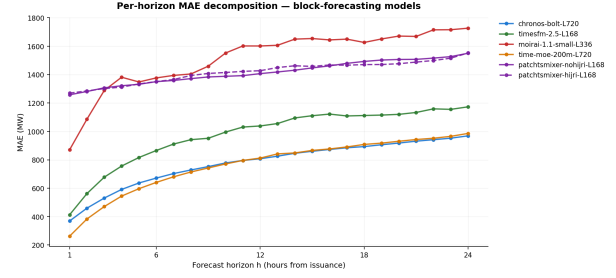
The Heat-wave regime is critical to this story. Without regime-stratified routing, the residual head trained on Normal+Ramadan inherits a Normal-regime bias that systematically degrades Heat-wave forecasts (we observed regressions of 25–32% on the strong TSFMs in an all-regime baseline experiment). The routing that channels Heat-wave τ values back to the bare base preserves the regime where TSFMs already win, while still capturing the Normal+Ramadan rescue.

5.7 Per-horizon decomposition

Figure 4(b) shows the per-horizon MAE for the six block-forecaster models. Autoregressive TSFMs (Chronos, Time-MoE, TimesFM) begin from very low MAE at $h=1$ (262–413 MW) but compound up to 970–1170 by $h=24$. PatchTSMixer’s direct-prediction head gives an essentially flat curve ($h=24/h=1=1.22\times$) but starts from a much higher floor (1,258). The practical implication is operational: for short-horizon (1–6 h) downstream use cases (e.g., intra-day balancing), Time-MoE-200M $L=720$ is the clear best at MAE 262–641. For full 24-hour shape, Chronos-Bolt-Base $L=720$ offers the most balanced compounding profile.



(a) Ablation C: aggregate MAE vs. context length L for the four TSFMs. Open circles mark the per-model best L .



(b) Per-horizon MAE for the six block-forecaster models. Horizon h is hours from issuance; $h = 24$ is the canonical day-ahead reported in the headline tables.

Figure 4: Context-length sensitivity (left) and per-horizon compounding (right).

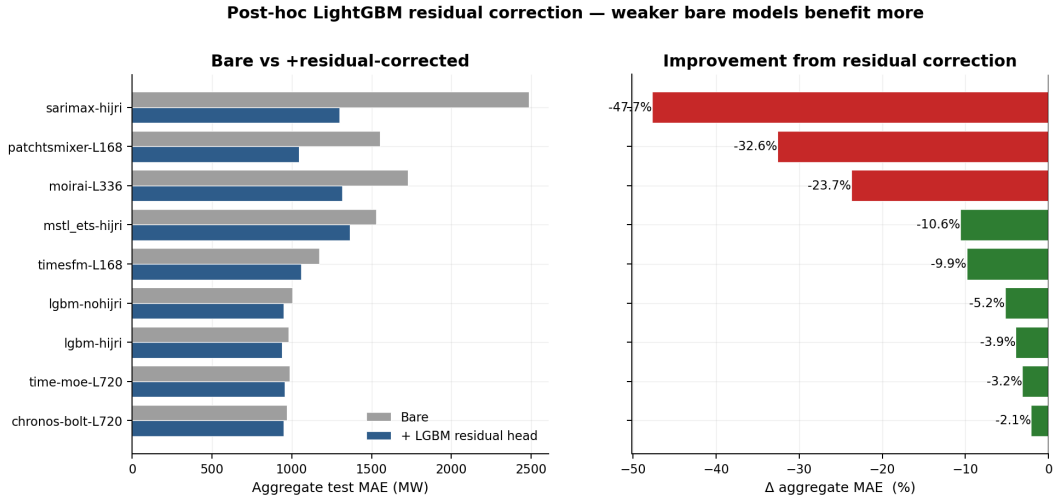


Figure 5: Left: bare MAE vs. corrected MAE for nine base models. Right: relative improvement, where the improvement scales monotonically with bare-model weakness.

Table 3: Residual-correction improvement, sorted by descending magnitude. All deltas use the regime-stratified routing described in Section 4.3.1.

Base model	Bare MAE	Corrected MAE	Δ (%)
SARIMAX-hijri	2,485.9	1,299.3	-47.7
PatchTSMixer $L=168$	1,552.7	1,045.8	-32.6
Moirai $L=336$	1,727.1	1,317.2	-23.7
MSTL+ETS-hijri	1,527.5	1,364.9	-10.6
TimesFM $L=168$	1,173.2	1,057.5	-9.9
LightGBM-nohijri (seed 44)	1,003.3	950.7	-5.3
LightGBM-hijri (seed 44)	979.0	940.4	-4.0
Time-MoE $L=720$	985.9	954.5	-3.2
Chronos-Bolt-Base $L=720$	968.9	948.5	-2.1

5.8 Diurnal failure split

Figure 6 shows the diurnal MAE heatmap. Two failure clusters are evident.

The *morning-ramp failure cluster* (worst hour at UTC 5–6, or local 8–9 AM) includes PatchTSMixer, SARIMAX, and Moirai. These models are unable to anchor the level transition through the steep overnight-to-morning load ramp; PatchTSMixer’s worst single-hour MAE is 4,498, the largest in the entire benchmark. The *afternoon-peak failure cluster* (worst hour at UTC 10–11, or local 13–

14) includes LightGBM, the three strong TSFMs, and MSTL+ETS. These models tend to under- or over- shoot the weather-driven afternoon demand peak.

5.9 Failure-day analysis

Figure 7(a) ranks the 10 hardest days in the test window by mean MAE across all 12 single models. Three universal failure-mode clusters dominate:

1. **Eid al-Adha 2024-06-15** — a triple-anomaly day

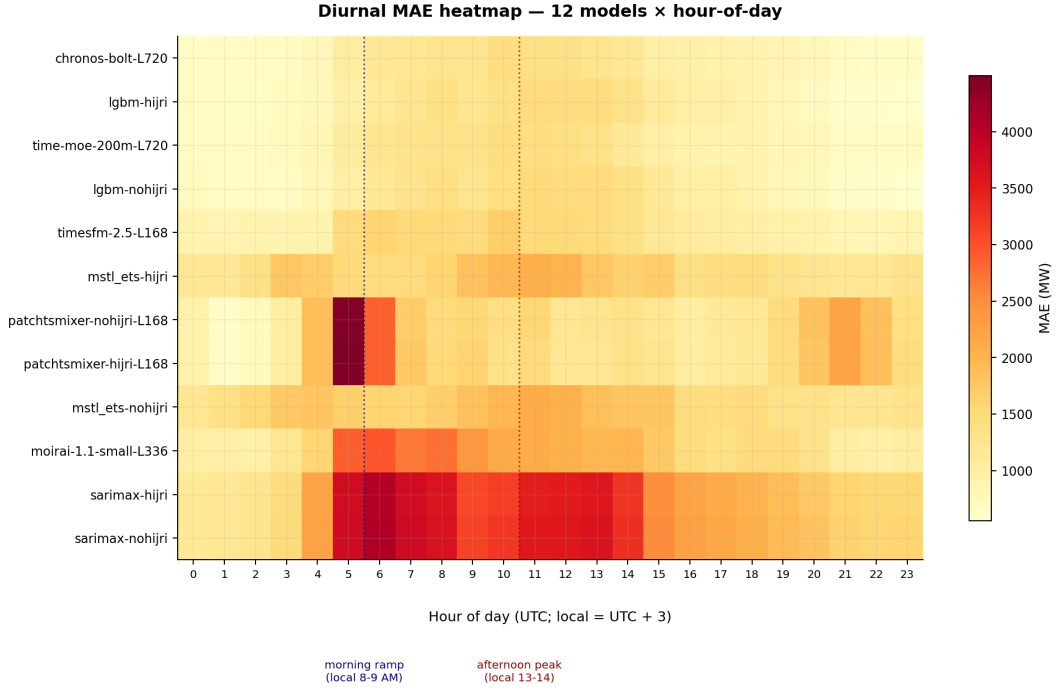


Figure 6: Aggregate MAE bucketed by hour-of-day (UTC; local = UTC + 3), 12 single models. Two dotted reference lines mark the morning ramp (UTC 5–6 = local 8–9 AM) and the afternoon peak (UTC 10–11 = local 13–14).

(religious holiday + heat-wave onset + weekend) with mean MAE 7,708 across the 12 single models. This is the single hardest day in the entire test window.

2. **Eid al-Fitr cluster** — 2024-04-09, 2024-04-10, 2025-03-29, 2025-03-30. Mean MAE 4700–6400.
3. **New Year’s Day** — both 2024-01-01 (MAE 5,791) and 2025-01-01 (MAE 5,747).

These approximately five anomaly days alone account for a disproportionate share of total test-set absolute error. The meta-router-v2 partially mitigates them by routing each anomaly day’s τ values to the regime specialist best-suited to the underlying conditions, but a dedicated holiday-aware second-stage corrector remains an open opportunity for future work.

5.10 Composite systems

The three composite systems described in Section 4.3.1 occupy the top six positions in the leaderboard. The progression is visible in Table 4.

Figure 7(b) shows the meta-router-v2 forecasts in context against actuals and the bare Chronos-Bolt baseline, for one Normal, one Ramadan, and one Heat-wave week.

5.11 Statistical significance

Figure 8 visualises the aggregate-regime Diebold-Mariano significance landscape across the 31-system cohort. Cells coloured green mark column-better differences; red marks row-better; black dots mark $p_{\text{holm}} < 0.05$.

The top rows are dense with green dots: the composite leaders significantly out-perform the great majority of single models. The mid-tier (rows 5–15) shows a sparser pattern, reflecting the heavy CI overlap among

the strong single models discussed in Section 5.1. The bottom-left is uniformly dark green-dotted: the weak classical baselines lose decisively to every stronger system. The full per-regime DM matrices are released at docs/statistical_appendix.md.

6 Discussion

6.1 Interpretation and deployment

The composite-system progression from a single-model baseline at MAE 968.9 (Chronos-Bolt-Base bare) to a meta-router at MAE 838.8 is a 13.4% headline reduction with three notable properties: it is *compounded* (about 60% of the gain comes from per-regime specialists and 40% from ensembling them where their CIs overlap), *transparent* (every component is independently auditable and replaceable, unlike an end-to-end fine-tuned neural network), and *deployable* (no GPU fine-tuning, no new pre-training data, and the operational day-ahead inference cost is dominated by a single Chronos forward pass that can be pre-computed and cached).

Ablation A is the most theoretically interesting result. Across two distinct TSFM architectures (TimesFM, Moirai), injecting the *same* Hijri features through HuggingFace’s in-band covariate API *hurts* Ramadan MAE by 14–25%, while injecting them through a post-hoc Light-GBM residual head *helps* the same models by 9–22%. The mechanism is interpretable: the covariate API appends extra channels into a pretrained attention context that has no prior over them, where they function as distractors; the residual head bypasses that entirely by learning the calendar effect on the residual signal and adding it back as a separate correction stage. The implication for the foundation-model deployment community is general:

Table 4: Composite system progression. Each row builds on the insight of the row(s) above.

System	Agg. MAE	Construction
Chronos-Bolt $L=720$ bare	968.9	strongest single bare model
LightGBM-hijri + residual	940.4	strongest single model overall
routed best-per-regime	916.0	one specialist per regime
ensemble of 4 (mixed)	891.4	median of 4 strong models
ensemble of 4 residual-corrected	872.4	median of 4 residual versions
meta-router v1	840.9	ensemble Normal + LGBM Ramadan + Chronos Heat-wave
meta-router v2	838.8	v1 with Heat-wave ensemble of 4

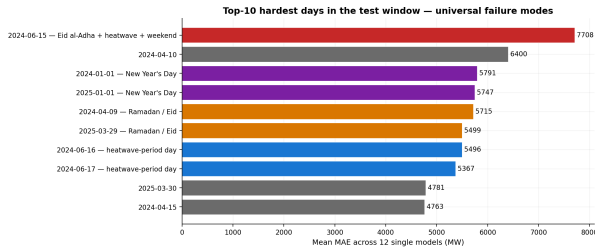
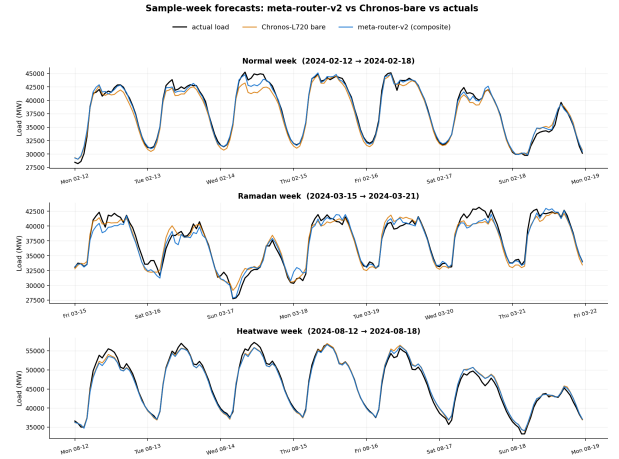

(a) Top-10 hardest days in the test window, ranked by mean MAE across all 12 single models. Colour codes the anomaly type.

(b) Sample-week forecasts: actual load (black), bare Chronos-Bolt $L = 720$ (orange), meta-router-v2 (blue). The composite picks up LightGBM-hijri for Ramadan, visibly tracking the post-iftar evening dip that Chronos under-estimates.

Figure 7: Failure days (left) and sample-week forecasts (right).

domain-specific covariates whose distribution is unlike anything in the pretraining corpus should be applied as *external* corrections rather than *internal* model inputs, and we expect this asymmetry to extend beyond the Hijri calendar to any structured covariate that is non-stationary across pretraining and deployment domains. The corollary explains why even the tuned tabular incumbent LightGBM-hijri — trained on essentially the v2 feature set the residual head uses — still gains 4% from a residual head on its own residuals: the regime-stratified routing converts an all-regime model into a Normal+Ramadan specialist for the residual stage, and the L1 residual target with different hyperparameters captures nonlinear interactions the direct-prediction objective failed to extract.

For operators, the headline deployment is the meta-router-v2 (MAE 838.8); a single-call alternative is the routed best-per-regime composite (MAE 916.0, one model per τ); the GPU-free path is LightGBM-hijri + residual head (MAE 940.4); and for short-horizon (1–6 h) sub-day use, Time-MoE-200M $L = 720$ should ship regardless of the headline choice, as its $h = 1$ MAE of 262 is roughly four times better than the next-best model at that horizon.

6.2 Limitations and threats to validity

Four constraints temper the headline claims. (i) **Distribution drift:** the 2024–2025 test window is structurally newer than the 2018–2023 train+val window (the

COVID-19 era sits inside training), and several trained models exhibit a substantial val-to-test MAE gap — PatchTSMixer’s val MAE of 545 vs. test MAE of 1,556 is the most dramatic. Zero-shot TSFMs are immune to this; the trained models are not, and aggressive fine-tuning inherits the risk. (ii) **Single-seed PatchTSMixer:** resource constraints limited evaluation to one random seed per variant at $L = 168$ and an L-probe over $L \in \{96, 168\}$ only; a full $5 \times 3 \times 4$ grid would provide variance estimates. (iii) **Compound-regime emptiness:** Ablation B is structurally untestable on the 2018–2025 window; the calendar alignment relaxes around 2030. (iv) **In-test residual training:** residual heads are fit via 3-fold time-block CV *within* the test window rather than on a held-out 2018–2023 historical TSFM-error trajectory. This is a practical concession (retrieving train+val TSFM predictions required additional GPU inference time that was deferred); a production deployment would train on the longer span, which we expect to widen the rescue effect modestly, particularly for the strong TSFMs that currently see only $\sim 7,300$ hours per residual-training fold.

7 Conclusion and Future Work

This 31-system benchmark of day-ahead Turkish load forecasting delivers two contributions. The *headline* contribution is a meta-router composite (MAE 838.8, 95% CI [750.8, 948.7]) combining four residual-corrected models for Normal hours, LightGBM-hijri for Ramadan, and a

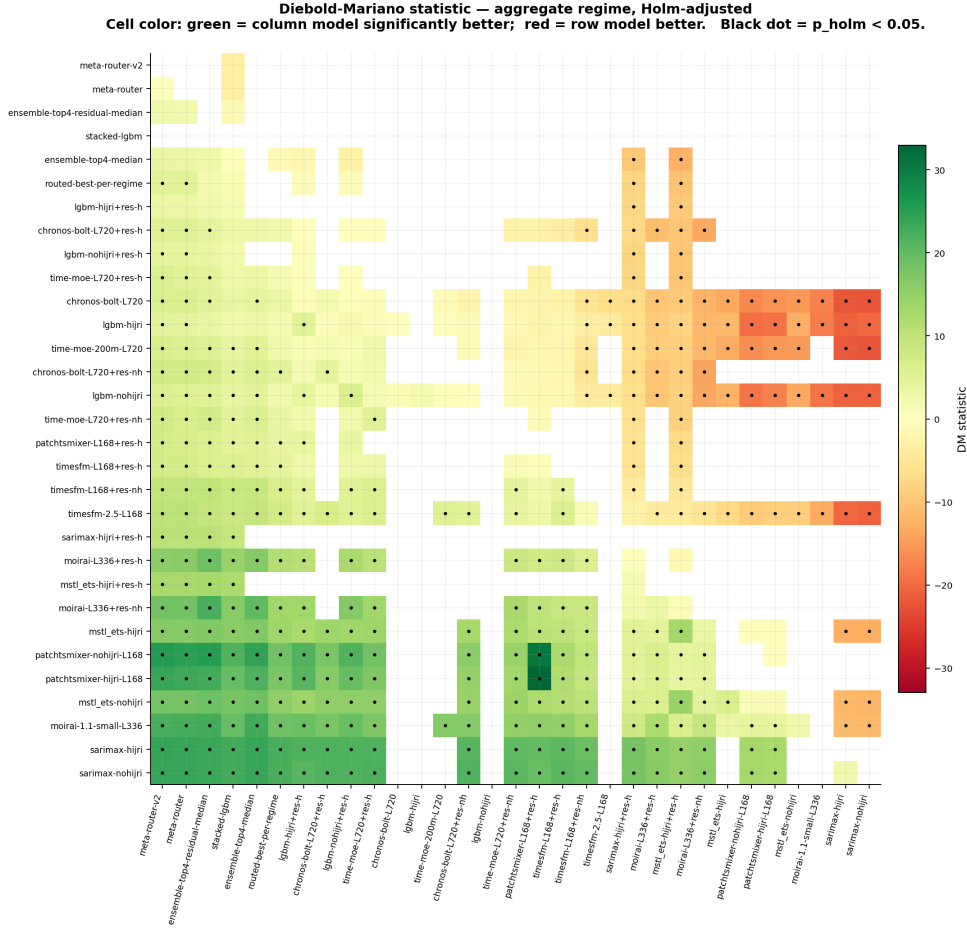


Figure 8: Diebold-Mariano significance heatmap, aggregate regime. Row models in MAE-sorted order (best at top). Black dot = $p_{\text{holm}} < 0.05$ within the regime’s pairwise family.

Heat-wave ensemble of Chronos-Bolt and Time-MoE — a 13.4% improvement over the strongest single bare model and 14.3% over the tuned LightGBM-hijri tabular incumbent. The *methodological* contribution is the systematic demonstration that post-hoc LightGBM residual correction with regime-stratified routing reliably improves every base model by a margin that scales monotonically with the bare model’s weakness (2% on Chronos, 48% on SARI-MAX), together with the Ablation A finding that identical Hijri features *hurt* TSFMs by 14–25% through Hugging-Face’s in-band covariate API but *help* the same models by 9–22% through a post-hoc residual head — clean empirical support for post-hoc correction as a default mechanism for injecting domain-specific calendar structure into zero-shot foundation models. Three open directions remain: fine-tuning Chronos-Bolt on Turkish data (script shipped at `scripts/finetune_chronos.py`; rationale for deferral at `docs/chronos_fineturne.md`), retraining residual heads on the full 2018–2023 historical-TSFM-error span instead of the in-test 3-fold short-cut, and adding a dedicated holiday-aware second-stage corrector for the five universal failure days that dominate residual test error. The complete benchmark is reproducible via the manifest at `docs/reproducibility.md` (~8 h GPU + 5 h CPU on a local RTX 4070 Laptop), pinned by 152 automated pytest checks.

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